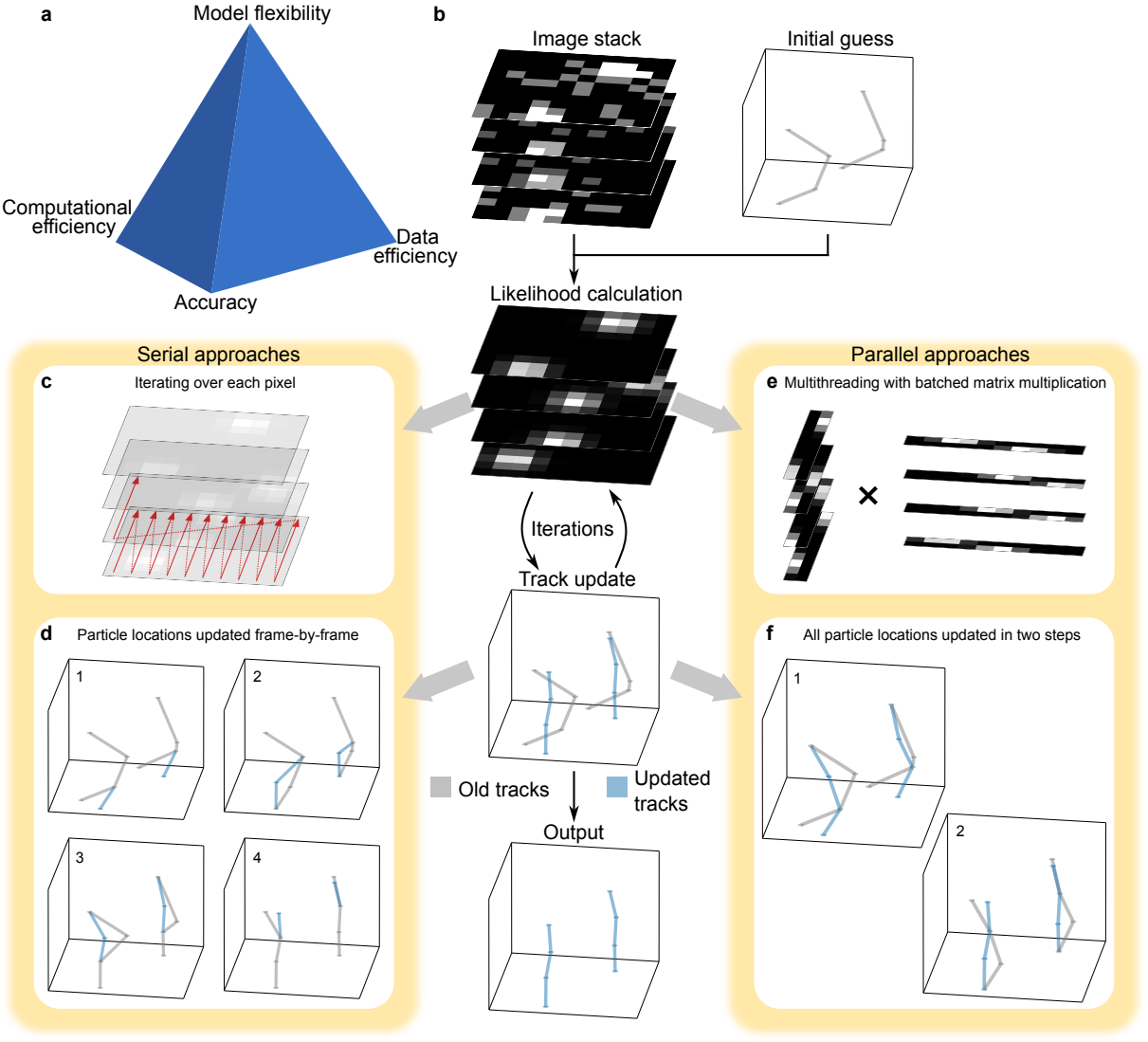

a
Model flexibility
Computational efficiency
Data efficiency
Accuracy
b
Image stack
Initial guess
Likelihood calculation
Serial approaches
c
Iterating over each pixel
Parallel approaches
e
Multithreading with batched matrix multiplication
Iterations
Track update
d
Particle locations updated frame-by-frame
1
2
3
4
Old tracks
Updated tracks
Output
f
All particle locations updated in two steps
1
2